

Lizzy Amaya:

From highest to lowest

#1 – we will not harm the turtles- In terms of "not harming" is not killing the turtle or hurting them in any way shape and form.

#2– Environment (cleanliness)- In order for the environment to stay clean, both turtles should be bathed and the tank should be cleaned constantly. (I don't know about CONSTANTLY, but perhaps weekly – that is more measurable).

#3– Basking (better heating and more room for the turtles)- the current tank is 24in by 24½ and is filled with about 65.5 gallons of water. I believe there is room to make bigger basking tank with the room available in the current tank. I The current basking platform is 15.5 x 13 x 10 inches so our group would estimate a 18.5 x 15 x 13 for the new basking platform we want to create.

#4 – Feeding (eating food)- obviously the turtles are required to eat in order to survive but since they are "overweight" (said by Attia) it isn't a higher problem, still very important but if they were to have food for a day I believe they would survive. (What counts as overweight? Do they have a measurement such as BMI for people?)

#5– Size of the tank (also amount of water in it)- we want to keep the same amount of water in the tank so this is why it's at the lowest part of the list about 66.5 gallons of water.

These design requirements ensure the turtle's needs are met (justifying):

Safety: Prevents injury and death (which we don't want), fulfilling moral and protection for the turtles

Cleanliness: Keeps the ecosystem of the tank balanced and reduces harmful bacteria that could lead to the turtles getting sick

Temperature: The temperature of the water in the tank keeps the turtle happy (not too cold or hot), and the temperature of the UV, which is 13 watts halogen and 5UV. This constant heat promotes strong shell growth

Feeding access: Ensures equal nutrition and prevents aggression

Tank size: Allows natural behaviors and long-term turtle growth

Stakeholder Solutions (how it fixes the problem of the stakeholder):

For the turtles: Promotes better health, longer lifespan, and comfort

For the caretakers and staff: Reduces cleaning time, prevents accidents, and ensures safe handling

For the students and researchers: Creates a stable, observable environment for educational use